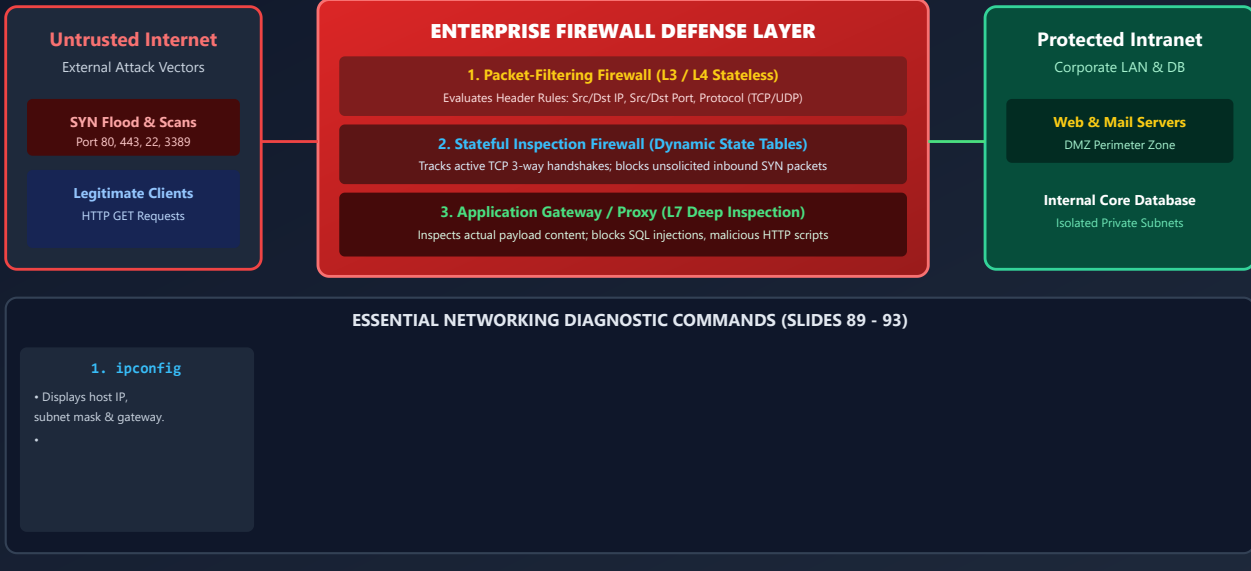


Module 13: Firewalls, Network Security & Diagnostic Commands

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FIREWALLS, NETWORK SECURITY & DIAGNOSTIC COMMANDS

Packet Filtering, Stateful Inspection, Application Gateways & Essential Utilities (ping, tracer, netstat)



/all shows MAC & DNS. • /flushdns clears cache. Linux: ifconfig / ip a 2. nslookup • "Name System Lookup". • Queries DNS servers for domain mappings. • Checks MX & A records. e.g., nslookup aktu.ac.in 3. tracer • "Trace Route" utility. • Increments IP TTL (1,2,3) to list intermediate hops. • Identifies network delays. Linux: traceroute 4. ping • Tests end-to-end link. • Sends ICMP Echo Request (Type 8) & waits Reply (0). • Measures packet loss & RTT. e.g., ping 8.8.8.8 5. netstat • "Network Statistics". • Displays active TCP links and listening ports. • -ano shows PIDs & IPs. Essential for port audits.

1. Firewall Defensive Taxonomy

A firewall secures network perimeters by analyzing inbound and outbound packet flows against institutional access policies.

Three Architectural Tiers

- 1. Stateless Packet Filters:** Fast, rule-based matching on L3/L4 headers.
- 2. Stateful Inspection:** Dynamically tracks TCP handshake states to prevent unauthorized external ingress.
- 3. Application Gateways:** Terminates connections and validates application payloads at Layer 7.

2. Five Core Network Diagnostic Utilities (Slides 89 - 93)

Utility	Primary Target	Key Command Syntax
ipconfig	Local adapter IP, subnet mask, default gateway, and DNS configuration	ipconfig /all , ipconfig /flushdns
nslookup	Interactive DNS resolution and zone record inspection	nslookup aktu.ac.in
tracert	Hop-by-hop router discovery and latency profiling using incremental IP TTL	tracert 8.8.8.8
ping	Host reachability and packet loss verification using ICMP Echo Request/Reply	ping aktu.ac.in
netstat	Audit of established TCP socket connections, open listening ports, and PIDs	netstat -ano